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Problems Output Debug Console Terminal Ports 21
Version 4.0
Programmer: Jane Smith

Welcome to SIMS
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Enter Student ID: 1234567
Enter Student Name: Alice Johnson
Enter Current GPA: 3.84

Course Grades
Enter grade for Course 1: 95
Enter grade for Course 2: 88
Enter grade for Course 3: 91
Enter grade for Course 4: 84
Enter grade for Course 5: 100

Student information saved to student_record.txt

Recovered Student Record
Student ID : 1234567
Student Name : Alice Johnson
Current GPA : 3.84
Academic Standing : Honors
-----
Course 1 : 95
Course 2 : 88
Course 3 : 91
Course 4 : 84
Course 5 : 100
-----
Does the recovered information match the original record? (Y/N): y

Verification Result: PASS

Student Summary
Student ID : 1234567
Student Name : Alice Johnson
Current GPA : 3.84
Academic Standing : Honors
-----
Course Grades
Course 1 : 95
Course 2 : 88
Course 3 : 91
Course 4 : 84
Course 5 : 100
-----
Average Grade : 91.60
Highest Grade : 100
Lowest Grade : 84
student@716d5f035fe:/workspace$
```

```
student_record.txt
File Edit View
H1
1234567
Alice Johnson
3.84
Honors
95
88
91
84
100

Ln 1, Col 1 50 characters Plain text 100% Unix (LF) UTF-8
```

Answers for customer:

1. What should the data file be named? `student_records.txt`, saved in the same folder as the program. This matches Example 3 and the submission requirements. I'd define it as a constant once, so it's easy to change later.
2. Should the program create the file if it doesn't exist? Yes. Opening the file in write mode ("w") creates it automatically, so the Registrar doesn't need to create it by hand.
3. Overwrite or append? Overwrite. This release stores only one student, so appending would leave stale records in the file and break the read-back.
4. What if the file can't be opened? Show the error message from Example 3 (naming the file and suggesting the user check that it exists and is accessible), skip the rest of the read/verify steps, and exit cleanly without crashing. This applies to both writing and reading.
5. Should the program automatically read the file after saving? Yes. Example 1 shows "Student information successfully saved," followed directly by "Reading student information...". No extra prompt is needed.
6. Should the recovered report exactly match the original report? The data should match exactly. The recovered report uses the same layout, with the heading "Recovered Student Record" as in the example. The average, highest, and lowest grades are recalculated from the five grades read in.
7. Should course names be saved, or only grades? Grades only. The Registrar's file layout in Example 2 contains no course names, and the requirements specify saving the five grades.
8. Recalculate academic standing after reading, or read it from the file? Read it from the file, since the requirement is to store it and retrieve it. For the verification step, compare the value read from the file against the value originally calculated in memory. If they differ, the file is corrupt or was edited.
9. What if the file has invalid or incomplete data? Check the return value of every read (`fscanf` or `fgets`). If something is missing or out of range (grades outside 0–100, GPA outside 0.0–4.0), display an error, don't show a partial report, and report that verification failed.
10. Should blank lines be included in the file? No. Use one value per line with no blank lines, exactly as in Example 2. This keeps reading simple and predictable. Because the student name contains spaces, read that line with `fgets` (and strip the newline) rather than `%s`.

11. Will future versions store multiple students in one file? Not in this release, since it's listed as out of scope. I'm assuming it's likely in the future, so the one-value-per-line layout could later be repeated per student.
12. Should employees be able to choose the filename? Not in this release. The filename is fixed. A future change request could add a filename prompt or command-line argument.